UNIT 2

ARTIFICIAL INTELLIGENCE

NEW SYLLABUS

UNIT 2

Advanced Concepts of Modeling in AI

Learning Objectives:

1. To familiarize students with supervised, unsupervised and reinforcement learning based approach.
2. To Introduce students to the neural network.

Learning Outcomes:

1. Understand supervised, unsupervised and reinforcement learning based approach.
2. Understand Neural Networks.

Pre-requisites: Essential understanding of Artificial Intelligence

Key-concepts:

1. Supervised, unsupervised and reinforcement learning based approach
2. Neural Networks

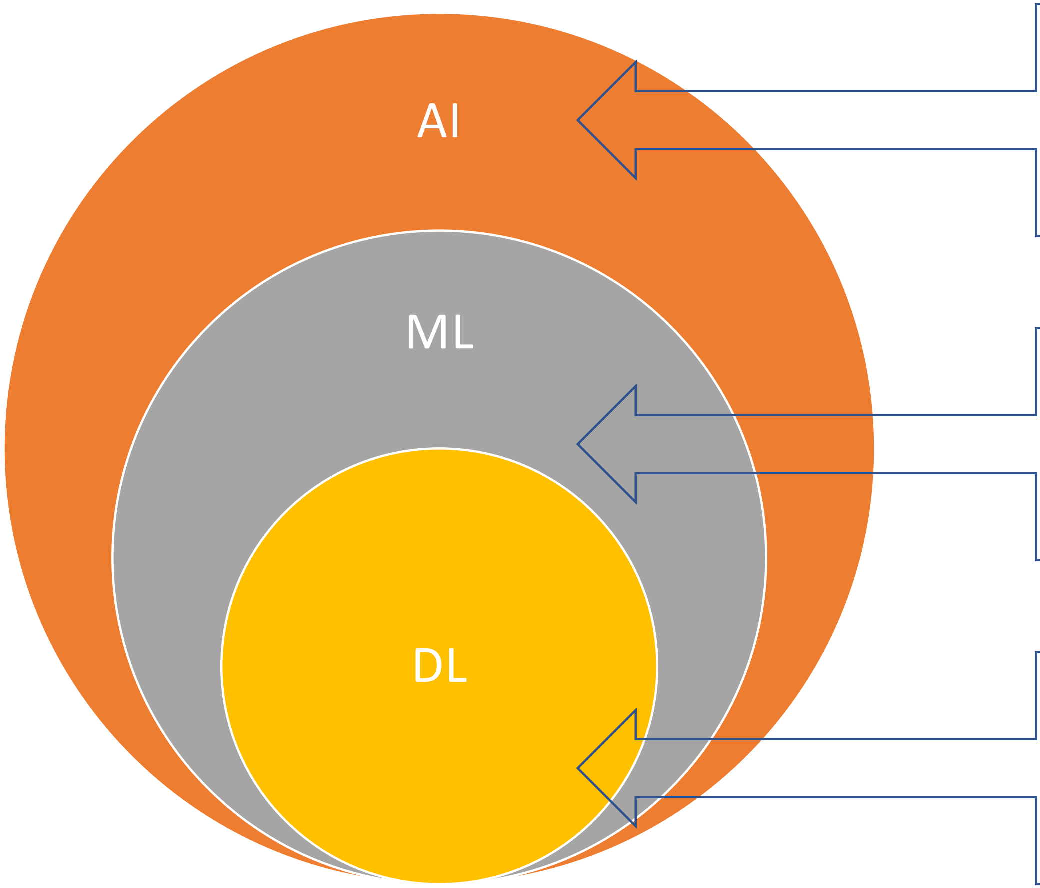
2.1 Revisiting AI, ML, DL

1 **Artificial Intelligence (AI)** is when computers are taught to think and learn like humans. They use information to help solve problems, answer questions, and do tasks.

2 **Machine Learning (ML)** helps computers learn from data and experience so they can do tasks better each time.

3 **Deep Learning (DL)** helps computers learn from lots of examples so they can do a task better and more accurately.

AI subset diagram



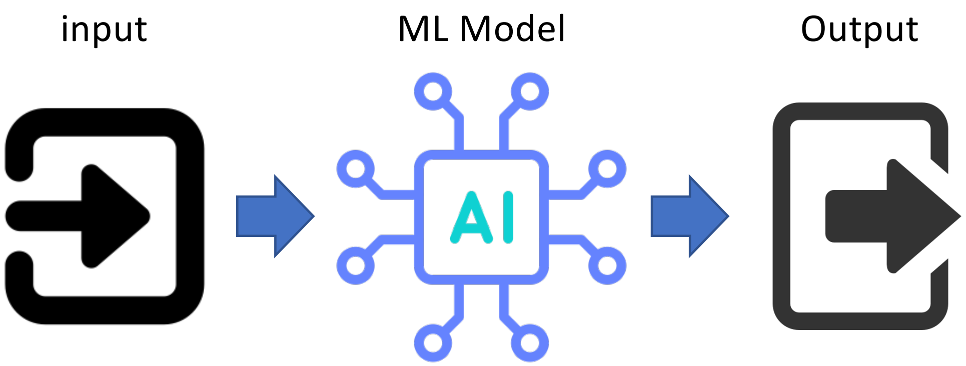
The ability of a machine to imitate human intelligence

ML is a subset of AI.
Algorithms that enables machines to learn from data and make decisions

DL is a subset of ML.
Algorithms inspired by the structure of a human brain (based on Artificial neural networks)

Machine Learning (ML)

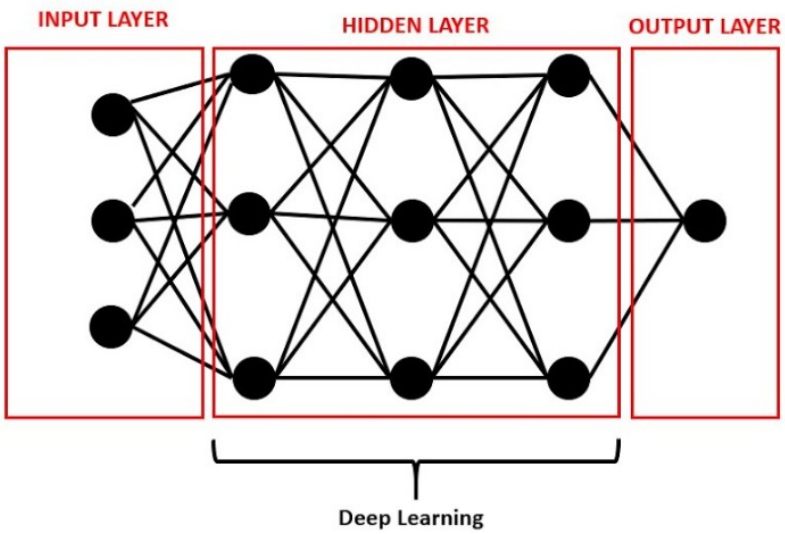
Machine Learning, or ML, enables machines to improve at tasks with experience. The machine learns from its mistakes and takes them into consideration in the next execution. It improvises itself using its own experiences.

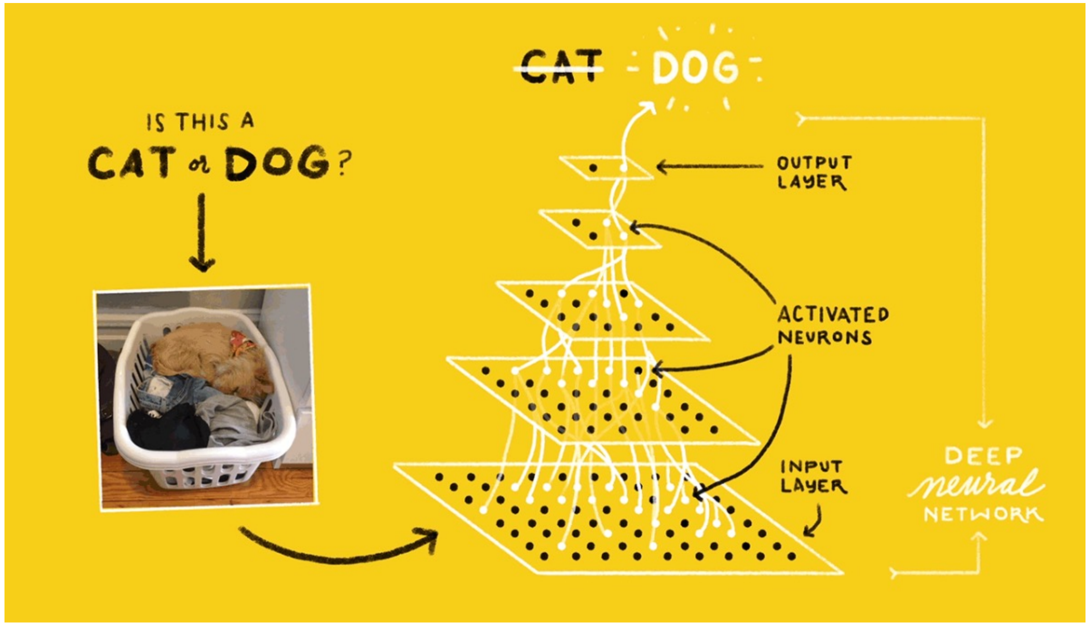


Deep Learning (DL)

Deep Learning, or DL, enables to train itself around the data to perform tasks with vast amounts of data. Deep Learning is the most advanced form of Artificial Intelligence.

ANN (Artificial Neural Network)





Common terminologies used with data

Fruit	Color	Price
Apple	Red	$1.8
Orange	Orange	$2
Banana	Yellow	$1
Graph	Purple	$3

What is Data ?

- <u>Data</u> is information in any form
- For example, A table with information about fruits is data
- Each row will contain information about the different fruits
- Each fruit is described by certain <u>features</u>

Common terminologies used with data

Features

Fruit	Color	Price
Apple	Red	$1.8
Orange	Orange	$2
Banana	Yellow	$1
Graph	Purple	$3

What do you mean by features?

- Columns of the tables are called features
- In the fruit dataset example, features may be name, color, price etc.,
- Some features are special, they are called labels

Common terminologies used with data

Label

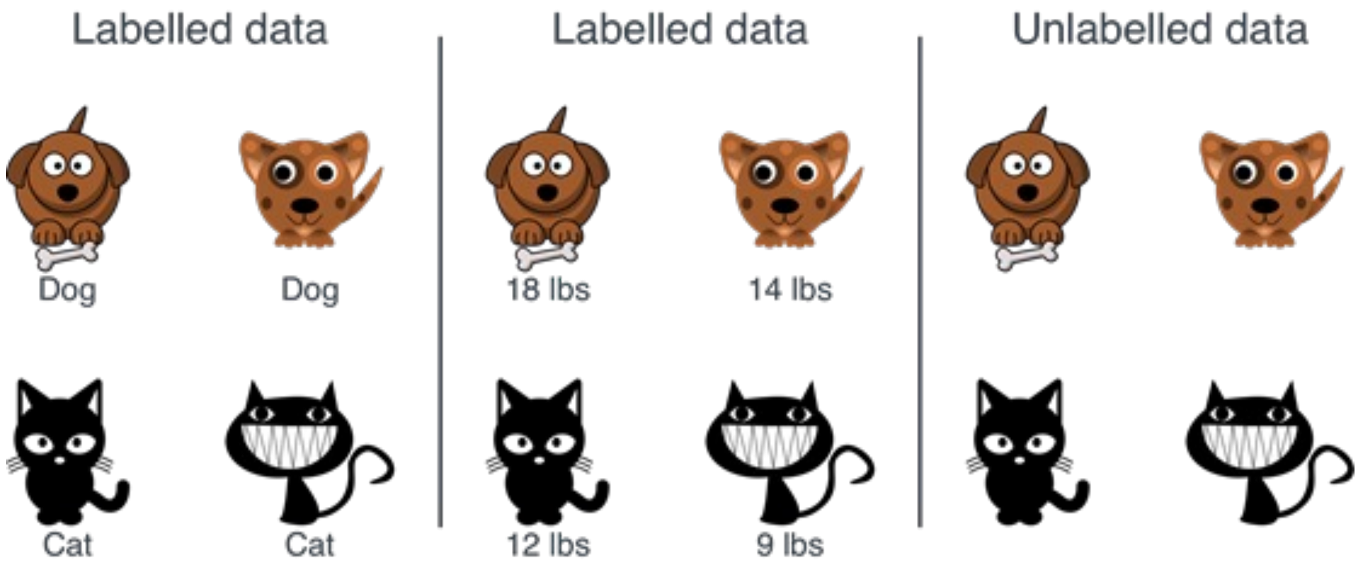
Features

Fruit	Color	Price
Apple	Red	$1.8
Orange	Orange	$2
Banana	Yellow	$1
Graph	Purple	$3

What are Labels?

- Data Labeling is the ***process of attaching meaning to data***
- It depends on the context of the problem we are trying to solve
- For example if we are trying to predict what fruit it is based on the color of the fruit, then color is the feature, and fruit name is the label.
- Data can be of two types – ***Labeled and Unlabeled***.

Common terminologies used with data

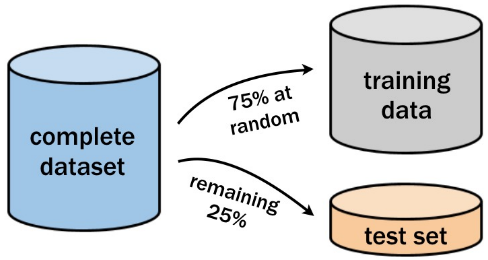


Labeled Data

- Data to which some tag/label is attached.
- For example name, type, number, etc.,

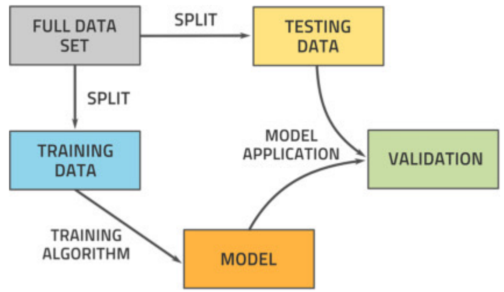
Unlabeled Data

- The raw form of data
- Data to which no tag is attached.



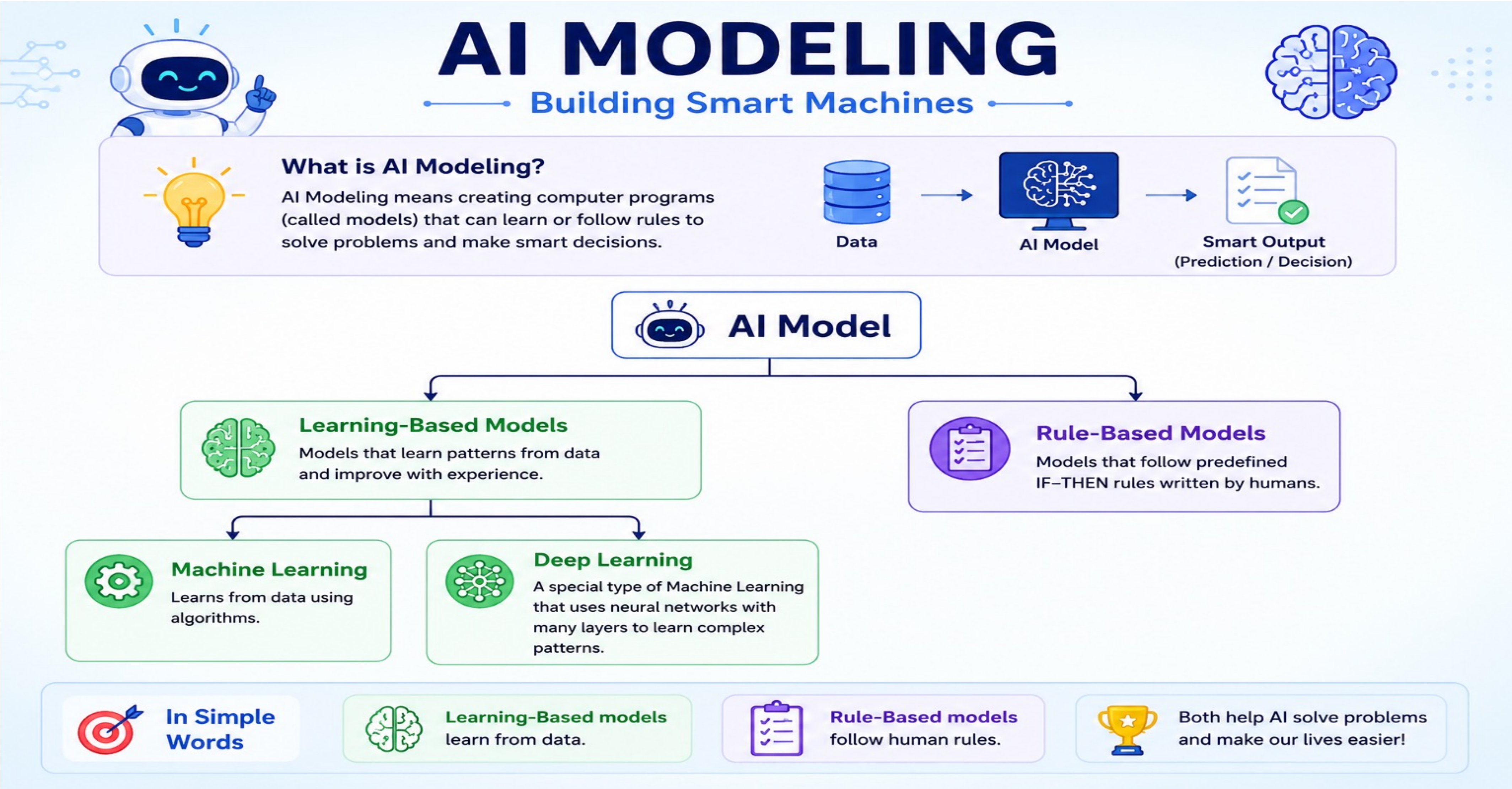
What do you mean by a <u>training</u> dataset?

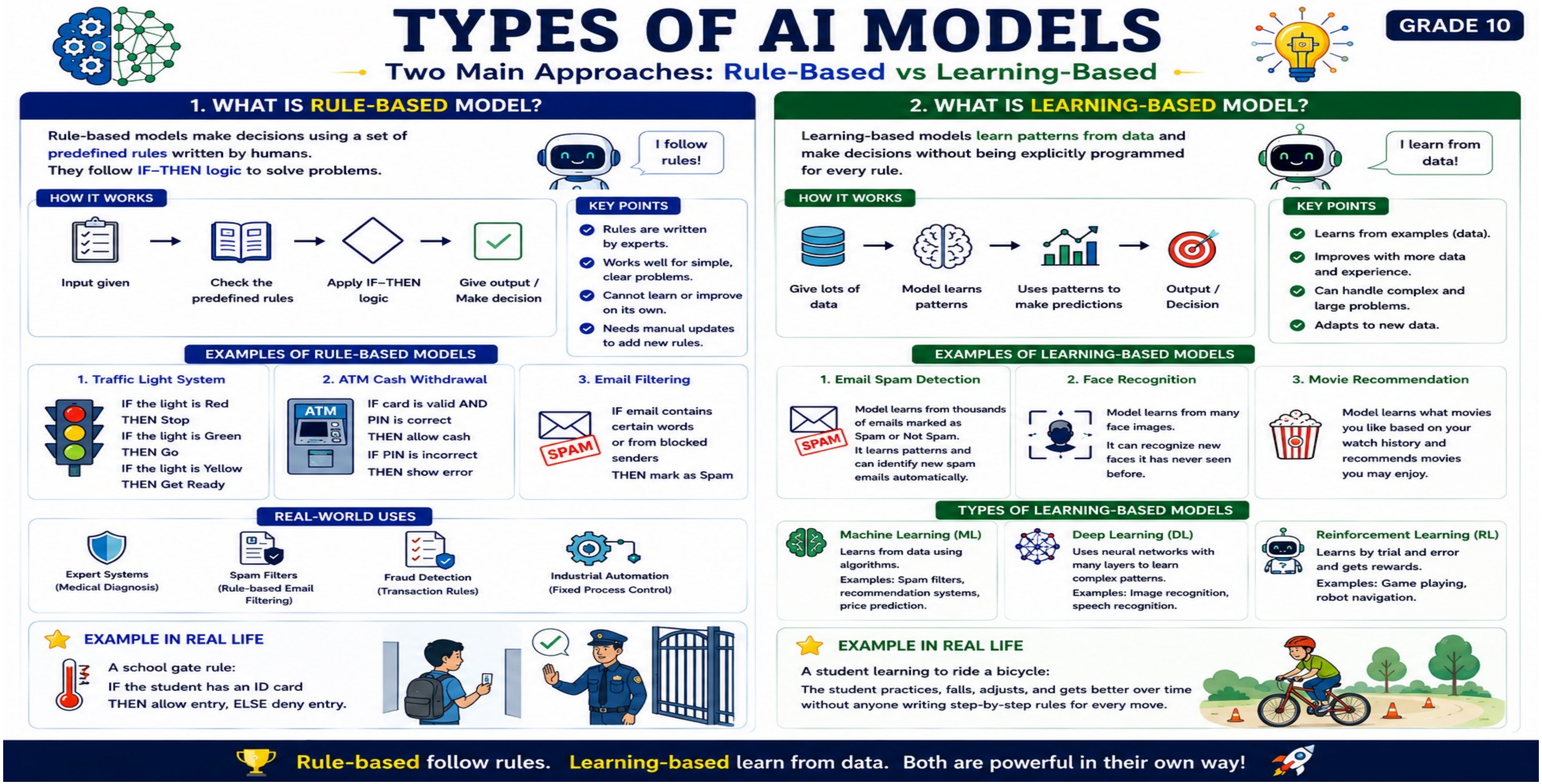
- The ***training dataset*** is a collection of examples given to the model to analyze and learn.
- Just like how a teacher teaches a topic to the class through a lot of examples and illustrations
- Similarly, a set of <u>labeled data</u> is used to train the AI model.

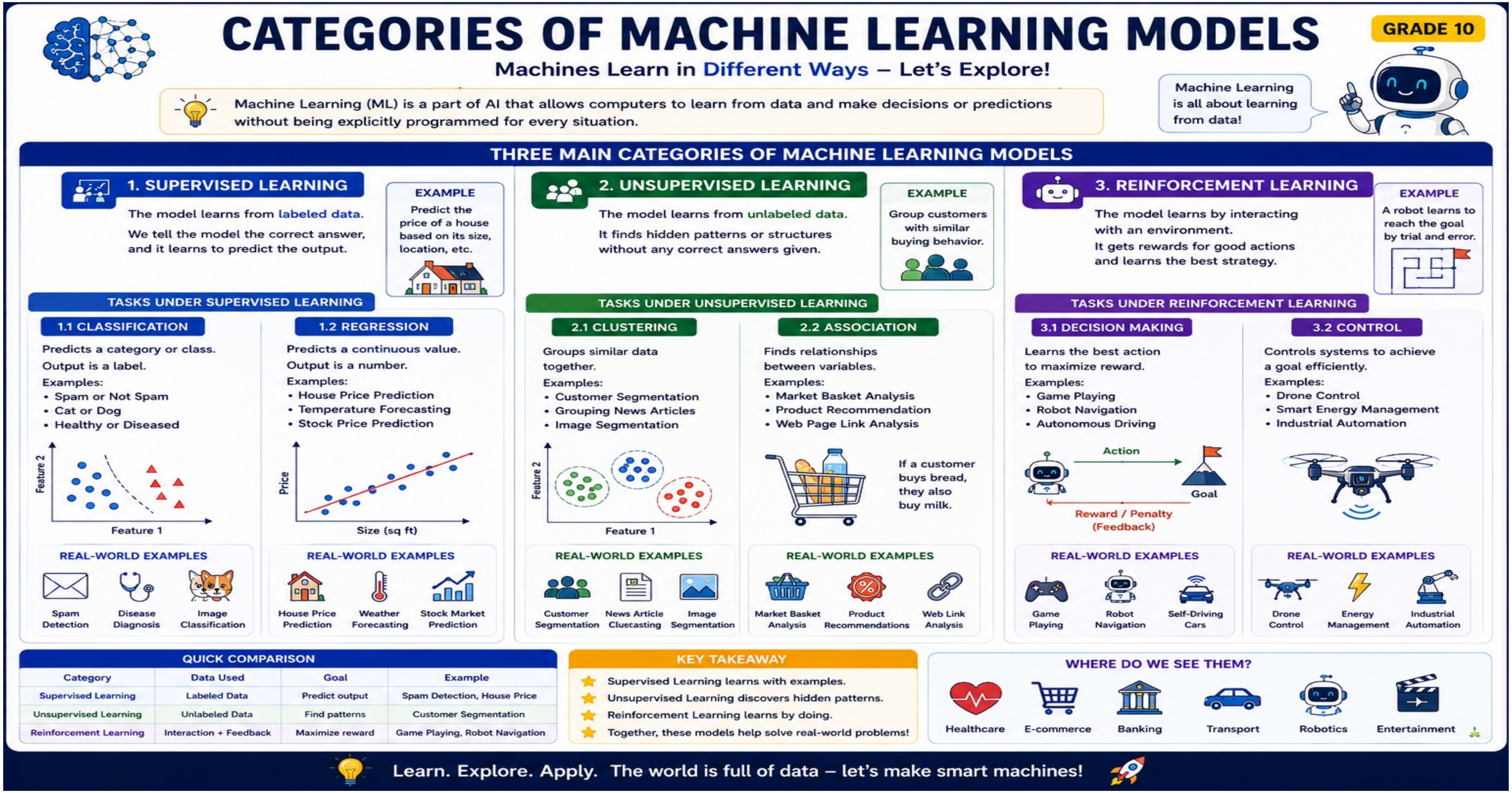


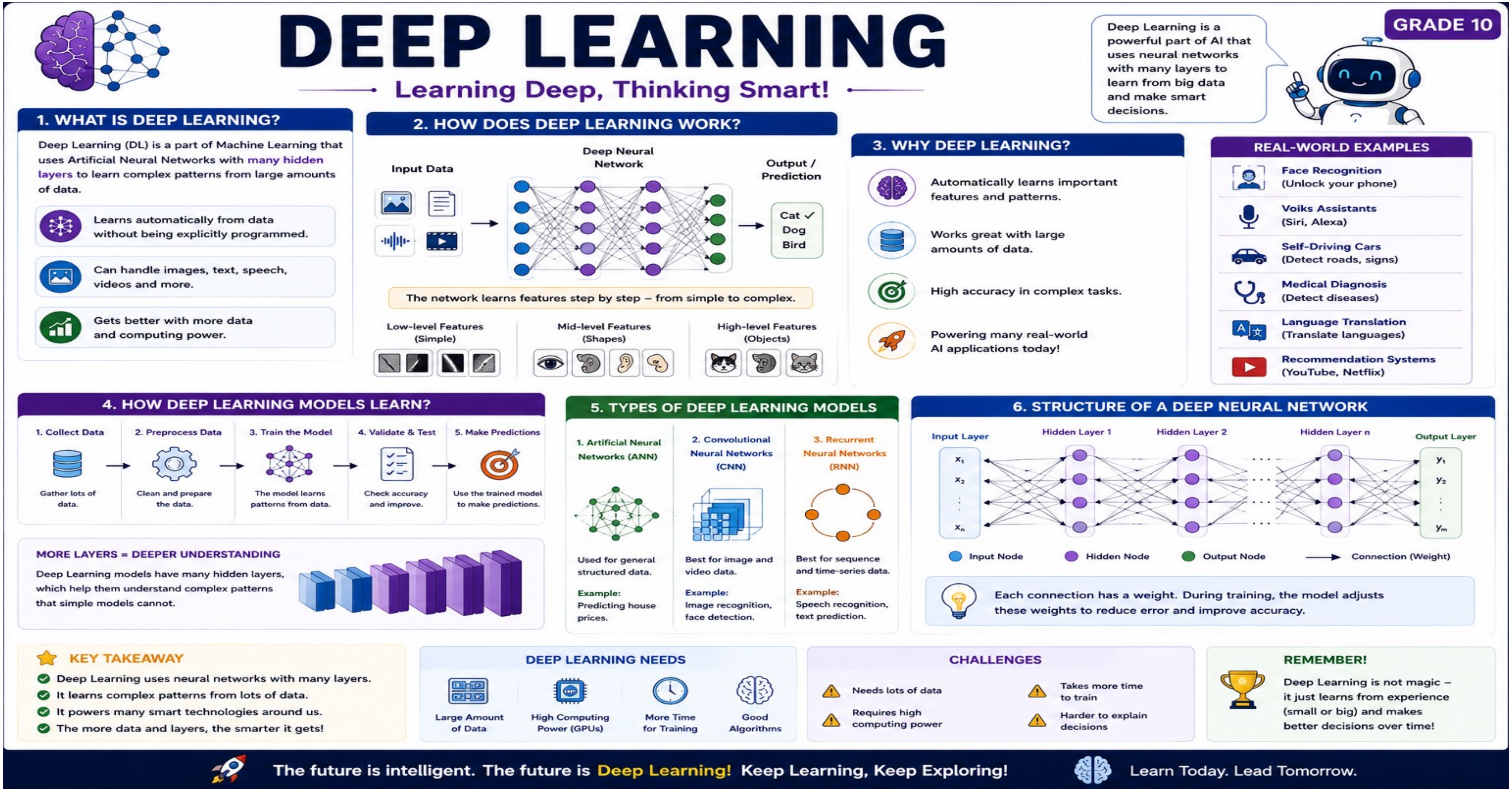
What do you mean by a <u>testing</u> dataset?

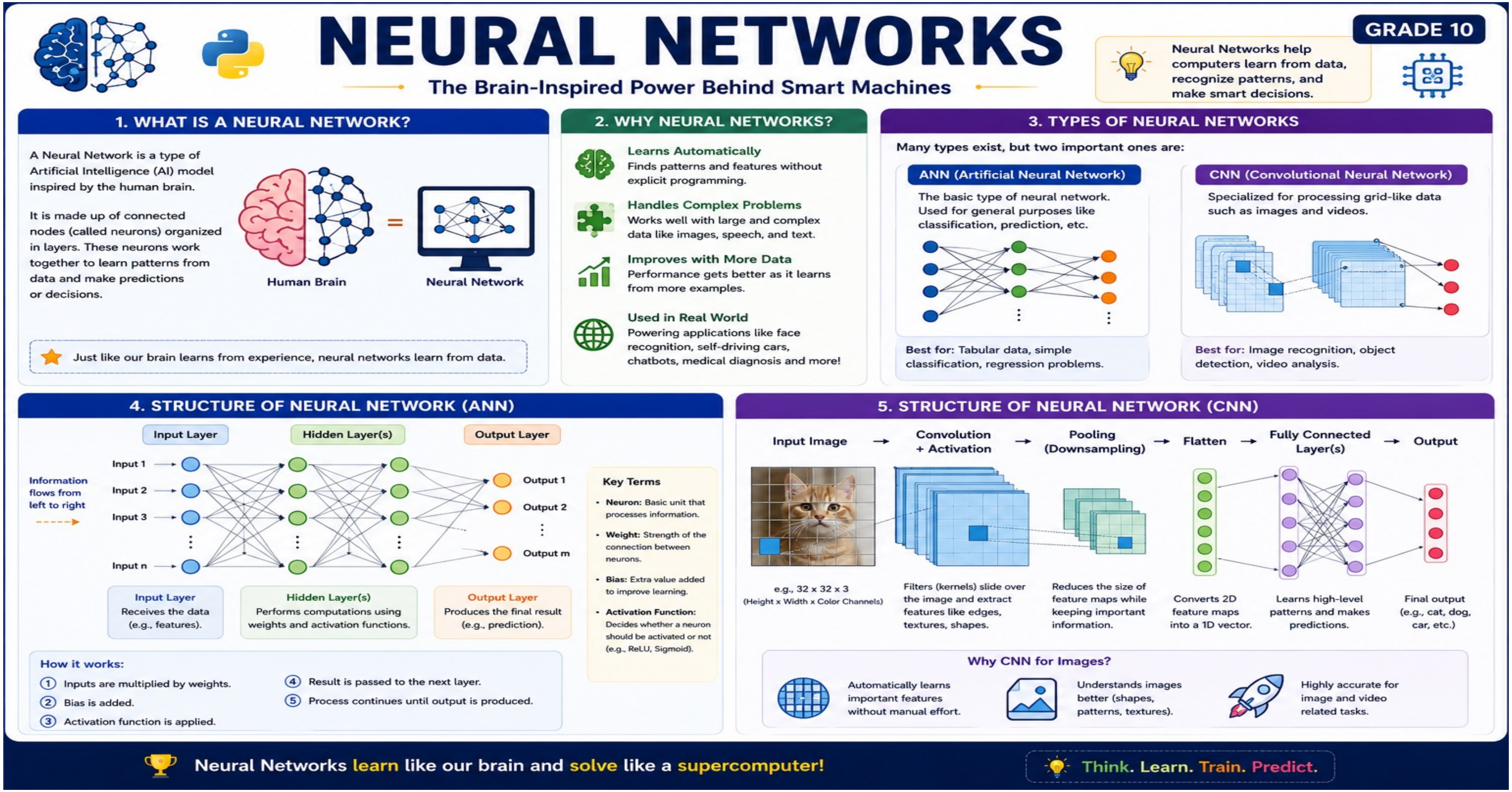
- The ***testing data*** set is used to test the accuracy of the model.
- Just like how a teacher takes a class test related to a topic to evaluate the understanding level of students.
- Test is performed <u>without labeled data</u> and then verify results with labels.


AI MODELING
Building Smart Machines
What is AI Modeling?
AI Modeling means creating computer programs (called models) that can learn or follow rules to solve problems and make smart decisions.
Data
AI Model
Smart Output
(Prediction / Decision)
AI Model
Learning-Based Models
Models that learn patterns from data and improve with experience.
Rule-Based Models
Models that follow predefined IF–THEN rules written by humans.
Machine Learning
Learns from data using algorithms.
Deep Learning
A special type of Machine Learning that uses neural networks with many layers to learn complex patterns.
In Simple Words
Learning-Based models learn from data.
Rule-Based models follow human rules.
Both help AI solve problems and make our lives easier!


TYPES OF AI MODELS
GRADE 10
Two Main Approaches: Rule-Based vs Learning-Based
1. WHAT IS RULE-BASED MODEL?
Rule-based models make decisions using a set of predefined rules written by humans.
They follow IF–THEN logic to solve problems.
I follow rules!
HOW IT WORKS
Input given
Check the predefined rules
Apply IF–THEN logic
Give output / Make decision
KEY POINTS
Rules are written by experts.
Works well for simple, clear problems.
Cannot learn or improve on its own.
Needs manual updates to add new rules.
EXAMPLES OF RULE-BASED MODELS
1. Traffic Light System
IF the light is Red THEN Stop
IF the light is Green THEN Go
IF the light is Yellow THEN Get Ready
2. ATM Cash Withdrawal
ATM
IF card is valid AND PIN is correct THEN allow cash
IF PIN is incorrect THEN show error
3. Email Filtering
SPAM
IF email contains certain words or from blocked senders THEN mark as Spam
REAL-WORLD USES
Expert Systems (Medical Diagnosis)
Spam Filters (Rule-based Email Filtering)
Fraud Detection (Transaction Rules)
Industrial Automation (Fixed Process Control)
EXAMPLE IN REAL LIFE
A school gate rule:
IF the student has an ID card
THEN allow entry, ELSE deny entry.
2. WHAT IS LEARNING-BASED MODEL?
Learning-based models learn patterns from data and make decisions without being explicitly programmed for every rule.
I learn from data!
HOW IT WORKS
Give lots of data
Model learns patterns
Uses patterns to make predictions
Output / Decision
KEY POINTS
Learns from examples (data).
Improves with more data and experience.
Can handle complex and large problems.
Adapts to new data.
EXAMPLES OF LEARNING-BASED MODELS
1. Email Spam Detection
SPAM
Model learns from thousands of emails marked as Spam or Not Spam.
It learns patterns and can identify new spam emails automatically.
2. Face Recognition
Model learns from many face images.
It can recognize new faces it has never seen before.
3. Movie Recommendation
Model learns what movies you like based on your watch history and recommends movies you may enjoy.
TYPES OF LEARNING-BASED MODELS
Machine Learning (ML)
Learns from data using algorithms.
Examples: Spam filters, recommendation systems, price prediction.
Deep Learning (DL)
Uses neural networks with many layers to learn complex patterns.
Examples: Image recognition, speech recognition.
Reinforcement Learning (RL)
Learns by trial and error and gets rewards.
Examples: Game playing, robot navigation.
EXAMPLE IN REAL LIFE
A student learning to ride a bicycle:
The student practices, falls, adjusts, and gets better over time without anyone writing step-by-step rules for every move.
Rule-based follow rules. Learning-based learn from data. Both are powerful in their own way!


CATEGORIES OF MACHINE LEARNING MODELS
GRADE 10
Machines Learn in Different Ways – Let's Explore!
Machine Learning (ML) is a part of AI that allows computers to learn from data and make decisions or predictions without being explicitly programmed for every situation.
Machine Learning is all about learning from data!
THREE MAIN CATEGORIES OF MACHINE LEARNING MODELS
1. SUPERVISED LEARNING
The model learns from labeled data.
We tell the model the correct answer, and it learns to predict the output.
EXAMPLE
Predict the price of a house based on its size, location, etc.
2. UNSUPERVISED LEARNING
The model learns from unlabeled data.
It finds hidden patterns or structures without any correct answers given.
EXAMPLE
Group customers with similar buying behavior.
3. REINFORCEMENT LEARNING
The model learns by interacting with an environment.
It gets rewards for good actions and learns the best strategy.
EXAMPLE
A robot learns to reach the goal by trial and error.
TASKS UNDER SUPERVISED LEARNING
1.1 CLASSIFICATION
Predicts a category or class.
Output is a label.
Examples:
• Spam or Not Spam
• Cat or Dog
• Healthy or Diseased
Feature 2
Feature 1
1.2 REGRESSION
Predicts a continuous value.
Output is a number.
Examples:
• House Price Prediction
• Temperature Forecasting
• Stock Price Prediction
Price
Size (sq ft)
REAL-WORLD EXAMPLES
Spam Detection
Disease Diagnosis
Image Classification
REAL-WORLD EXAMPLES
House Price Prediction
Weather Forecasting
Stock Market Prediction
TASKS UNDER UNSUPERVISED LEARNING
2.1 CLUSTERING
Groups similar data together.
Examples:
• Customer Segmentation
• Grouping News Articles
• Image Segmentation
Feature 2
Feature 1
2.2 ASSOCIATION
Finds relationships between variables.
Examples:
• Market Basket Analysis
• Product Recommendation
• Web Page Link Analysis
If a customer buys bread, they also buy milk.
REAL-WORLD EXAMPLES
Customer Segmentation
News Article Clustering
Image Segmentation
REAL-WORLD EXAMPLES
Market Basket Analysis
Product Recommendations
Web Link Analysis
TASKS UNDER REINFORCEMENT LEARNING
3.1 DECISION MAKING
Learns the best action to maximize reward.
Examples:
• Game Playing
• Robot Navigation
• Autonomous Driving
Action
Goal
Reward / Penalty (Feedback)
3.2 CONTROL
Controls systems to achieve a goal efficiently.
Examples:
• Drone Control
• Smart Energy Management
• Industrial Automation
REAL-WORLD EXAMPLES
Game Playing
Robot Navigation
Self-Driving Cars
REAL-WORLD EXAMPLES
Drone Control
Energy Management
Industrial Automation
QUICK COMPARISON
Category | Data Used | Goal | Example
Supervised Learning | Labeled Data | Predict output | Spam Detection, House Price
Unsupervised Learning | Unlabeled Data | Find patterns | Customer Segmentation
Reinforcement Learning | Interaction + Feedback | Maximize reward | Game Playing, Robot Navigation
KEY TAKEAWAY
Supervised Learning learns with examples.
Unsupervised Learning discovers hidden patterns.
Reinforcement Learning learns by doing.
Together, these models help solve real-world problems!
WHERE DO WE SEE THEM?
Healthcare
E-commerce
Banking
Transport
Robotics
Entertainment
Learn. Explore. Apply. The world is full of data – let's make smart machines!


DEEP LEARNING
Learning Deep, Thinking Smart!
Deep Learning is a powerful part of AI that uses neural networks with many layers to learn from big data and make smart decisions.
GRADE 10
1. WHAT IS DEEP LEARNING?
Deep Learning (DL) is a part of Machine Learning that uses Artificial Neural Networks with many hidden layers to learn complex patterns from large amounts of data.
Learns automatically from data without being explicitly programmed.
Can handle images, text, speech, videos and more.
Gets better with more data and computing power.
2. HOW DOES DEEP LEARNING WORK?
Input Data
Deep Neural Network
Output / Prediction
Cat ✓
Dog
Bird
The network learns features step by step – from simple to complex.
Low-level Features (Simple)
Mid-level Features (Shapes)
High-level Features (Objects)
3. WHY DEEP LEARNING?
Automatically learns important features and patterns.
Works great with large amounts of data.
High accuracy in complex tasks.
Powering many real-world AI applications today!
REAL-WORLD EXAMPLES
Face Recognition (Unlock your phone)
Voiks Assistants (Siri, Alexa)
Self-Driving Cars (Detect roads, signs)
Medical Diagnosis (Detect diseases)
Language Translation (Translate languages)
Recommendation Systems (YouTube, Netflix)
4. HOW DEEP LEARNING MODELS LEARN?
1. Collect Data
Gather lots of data.
2. Preprocess Data
Clean and prepare the data.
3. Train the Model
The model learns patterns from data.
4. Validate & Test
Check accuracy and improve.
5. Make Predictions
Use the trained model to make predictions.
MORE LAYERS = DEEPER UNDERSTANDING
Deep Learning models have many hidden layers, which help them understand complex patterns that simple models cannot.
5. TYPES OF DEEP LEARNING MODELS
1. Artificial Neural Networks (ANN)
Used for general structured data.
Example: Predicting house prices.
2. Convolutional Neural Networks (CNN)
Best for image and video data.
Example: Image recognition, face detection.
3. Recurrent Neural Networks (RNN)
Best for sequence and time-series data.
Example: Speech recognition, text prediction.
6. STRUCTURE OF A DEEP NEURAL NETWORK
Input Layer
Hidden Layer 1
Hidden Layer 2
Hidden Layer n
Output Layer
Input Node
Hidden Node
Output Node
Connection (Weight)
Each connection has a weight. During training, the model adjusts these weights to reduce error and improve accuracy.
KEY TAKEAWAY
Deep Learning uses neural networks with many layers.
It learns complex patterns from lots of data.
It powers many smart technologies around us.
The more data and layers, the smarter it gets!
DEEP LEARNING NEEDS
Large Amount of Data
High Computing Power (GPUs)
More Time for Training
Good Algorithms
CHALLENGES
Needs lots of data
Takes more time to train
Requires high computing power
Harder to explain decisions
REMEMBER!
Deep Learning is not magic – it just learns from experience (small or big) and makes better decisions over time!
The future is intelligent. The future is Deep Learning! Keep Learning, Keep Exploring!
Learn Today. Lead Tomorrow.


NEURAL NETWORKS
The Brain-Inspired Power Behind Smart Machines
GRADE 10
Neural Networks help computers learn from data, recognize patterns, and make smart decisions.
1. WHAT IS A NEURAL NETWORK?
A Neural Network is a type of Artificial Intelligence (AI) model inspired by the human brain.
It is made up of connected nodes (called neurons) organized in layers. These neurons work together to learn patterns from data and make predictions or decisions.
Human Brain
Neural Network
Just like our brain learns from experience, neural networks learn from data.
2. WHY NEURAL NETWORKS?
Learns Automatically
Finds patterns and features without explicit programming.
Handles Complex Problems
Works well with large and complex data like images, speech, and text.
Improves with More Data
Performance gets better as it learns from more examples.
Used in Real World
Powering applications like face recognition, self-driving cars, chatbots, medical diagnosis and more!
3. TYPES OF NEURAL NETWORKS
Many types exist, but two important ones are:
ANN (Artificial Neural Network)
The basic type of neural network. Used for general purposes like classification, prediction, etc.
Best for: Tabular data, simple classification, regression problems.
CNN (Convolutional Neural Network)
Specialized for processing grid-like data such as images and videos.
Best for: Image recognition, object detection, video analysis.
4. STRUCTURE OF NEURAL NETWORK (ANN)
Input Layer
Hidden Layer(s)
Output Layer
Information flows from left to right
Input 1
Input 2
Input 3
Input n
Output 1
Output 2
Output m
Input Layer
Receives the data (e.g., features).
Hidden Layer(s)
Performs computations using weights and activation functions.
Output Layer
Produces the final result (e.g., prediction).
Key Terms
• Neuron: Basic unit that processes information.
• Weight: Strength of the connection between neurons.
• Bias: Extra value added to improve learning.
• Activation Function: Decides whether a neuron should be activated or not (e.g., ReLU, Sigmoid).
How it works:
1 Inputs are multiplied by weights.
2 Bias is added.
3 Activation function is applied.
4 Result is passed to the next layer.
5 Process continues until output is produced.
5. STRUCTURE OF NEURAL NETWORK (CNN)
Input Image → Convolution + Activation → Pooling (Downsampling) → Flatten → Fully Connected Layer(s) → Output
e.g., 32 x 32 x 3
(Height x Width x Color Channels)
Filters (kernels) slide over the image and extract features like edges, textures, shapes.
Reduces the size of feature maps while keeping important information.
Converts 2D feature maps into a 1D vector.
Learns high-level patterns and makes predictions.
Final output (e.g., cat, dog, car, etc.)
Why CNN for Images?
Automatically learns important features without manual effort.
Understands images better (shapes, patterns, textures).
Highly accurate for image and video related tasks.
Neural Networks learn like our brain and solve like a supercomputer!
Think. Learn. Train. Predict.

PERCEPTRON: The Building Block of AI Decisions

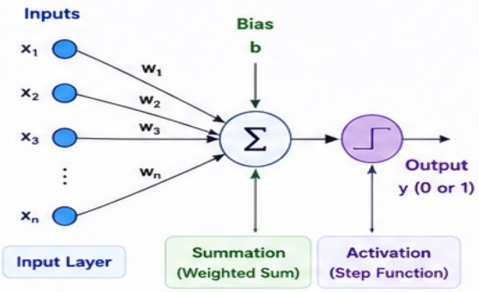
GRADE 10

A simple model that helps AI think and decide!

1. WHAT IS PERCEPTRON?

A Perceptron is the simplest type of Artificial Neural Network (ANN) model, proposed by **Frank Rosenblatt in 1958**.

- It is used for binary classification (two possible outputs: 0 or 1).
- It takes multiple inputs, does some calculations, and gives one output.
- It learns from examples and improves over time.



Perceptron = Weighted Sum of Inputs + Bias → Activation Function → Output (0 or 1)

2. HOW DOES AI MAKE DECISIONS WITH THE HELP OF PERCEPTRON?

STEP 1 Take Inputs	STEP 2 Calculate Weighted Sum	STEP 3 Add Bias	STEP 4 Apply Activation Function	STEP 5 Output Decision
The perceptron receives input values.	Multiply each input with its **weight** and add them.	Add the **bias** (b) to the result.	Apply the **step** activation function.	The perceptron gives output: 1 or 0
$x_1, x_2, \vdots, x_n$ →	$z = (x_1 \times w_1) + (x_2 \times w_2) + \dots + (x_n \times w_n)$ →	$z = z + b$ →	$y = \begin{cases} 1, & \text{if } z \ge 0 \\ 0, & \text{if } z < 0 \end{cases}$ →	$y = 1$ (Yes / True) or $y = 0$ (No / False)

If the weighted sum (plus bias) is positive or zero, output is 1.
If it is negative, output is 0.
That's how the perceptron makes a decision!

3. EXAMPLE: PERCEPTRON FOR EMAIL SPAM DETECTION

Goal: Decide whether an email is Spam (1) or Not Spam (0) based on some features.

Inputs (Features)

x_1	Contains word "free"	(1 = Yes, 0 = No)
x_2	Contains word "win"	(1 = Yes, 0 = No)
x_3	From unknown sender	(1 = Yes, 0 = No)

Weights and Bias (learned from examples)

$w_1 = 1.5$ $w_2 = 1.0$ $w_3 = 1.2$ $b = -2.5$

Example 1: Email A

Contains "free" and "win", unknown sender

Inputs: $x_1 = 1, \quad x_2 = 1, \quad x_3 = 1$

$$z = (1 \times 1.5) + (1 \times 1.0) + (1 \times 1.2) + (-2.5)$$
$$= 1.5 + 1.0 + 1.2 - 2.5$$
$$= 1.2$$

SPAM

Since $z \ge 0$ → Output $y = 1$ (Spam)

Decision: This email is Spam **Spam**

Example 2: Email B

Does not contain those words, known sender

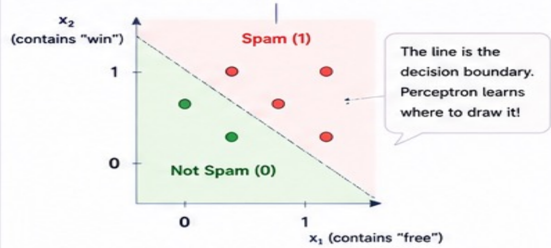
Inputs: $x_1 = 0, \quad x_2 = 0, \quad x_3 = 0$

$$z = (0 \times 1.5) + (0 \times 1.0) + (0 \times 1.2) + (-2.5)$$
$$= 0 + 0 + 0 - 2.5$$
$$= -2.5$$

Since $z < 0$ → Output $y = 0$ (Not Spam)

Decision: This email is Not Spam **Not Spam**

How Perceptron Separates Two Classes



KEY TAKEAWAY

Perceptron is a simple but powerful model that helps AI learn from data and make decisions between two choices.

REAL WORLD USES

- Spam Detection
- Medical Diagnosis (Yes/No)
- Fraud Detection
- Image Classification (Simple)
- Quality Control (OK / Not OK)

IN SHORT

Perceptron takes inputs, does math, applies a simple rule, and gives a decision — Yes (1) or No (0).
It's the first step toward building smart AI systems!

Simple today. Intelligent tomorrow. Learn. Build. Innovate.

AI begins with simple ideas!


THANK YOU
THANK YOU